

developer, you need to have a local server system set up to run your web application on your own computer before you unleash them onto the world. Setting up a local server is not only simple, it's also free!

For open source applications, typically you'd do your development in PHP and run it on an Apache server with the PHP module. Here, the Apache server is responsible for serving the content from your hard disk to the internet, and making it accessible. PHP does the processing bit, crunching numbers, making the web site more than just a collection of linked pages. Apache is then the HTTP web server with PHP being the server-side scripting language.

Most web sites will also require a data back end, to store anything that may need to be retrieved later on. Content management systems (CMSs) store the articles and user registrations; forums store user registrations and forum posts in the database, etc. The open source database server of choice is usually MySQL, where SQL stands for Structured Query Language. Here the data is accessed and modified by using commands written in the SQL syntax, which allows for complex modification of data in the database using simple English-like statements.

Setting these three up to work with each other can often be quite a pain for a n00b. To make things simple, there are many packages out there which set up all three for you in a single install, and configure them to work with each other, without requiring you to do any manual configuration. These packages known as AMP or Apache MySQL PHP packages are also free and open source, as are their individual components.

Note that an Apache Server running PHP with a MySQL database is not the only open source option. You will find a wide variety of free open source servers such as 'lighttpd' which as its name suggests is a lighter

server in use by web sites such as YouTube, **mininova.org**, **sourceforge.net**, and even Wikipedia. Many variants of Apache are also available since it is open source.

Instead of PHP one can use Perl, Python, Ruby, even JavaScript. However, PHP remains the most popular choice. Many web applications do exist which use alternatives to PHP. For example, Bricolage a CMS; Movable Type, a blogging platform; and TWiki, a wiki application all use Perl, some CMSs such as Plone and Django use Python. You are by no means limited. Since Perl, Python and Ruby are all open source, you can go as far as to develop your own custom version to suit your needs!

Databases also have a plethora of choices available besides MySQL. PostgreSQL is another popular alternative, which offers more features at a performance cost. Even besides these two popular ones there are many, many more which are perhaps not as commonly used. MySQL is the most popular open source database you will find, and it is extensively supported by most if not all web applications. Most are only compatible with MySQL and as such will not work on other variants, unless they offer a feature and usage parity. Of course, if you want to go far enough, you're a developer right? You can always make your own plugin!

Managing content

For almost anything that you want to develop, you are sure to find a framework or application that almost, if not completely, fits your needs. Looking to create a web site for creating and sharing documents, try

'opengoo'. Want to make a community social web site check out 'Elgg'. These are some of the more obscure example of web applications, one of the most common uses

is content management.

Content Management Systems or CMSs come with support for extensions in the form of plug-ins or modules, any kind of add-ons really. When developing a web site that deals with large amounts of content, it is impossible

to simply keep uploading connected HTML documents. You need to use a CMS that can categorise and tag the content, and present it to the user in a way that presents a clear way of navigation. The default configuration that these come in is unlikely to be suitable

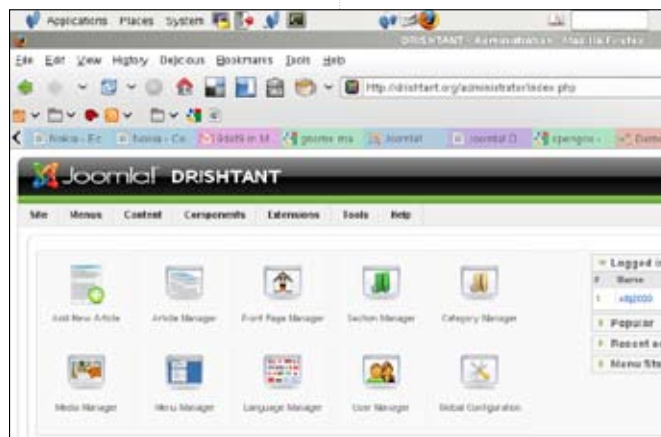
Joomla

This is a very popular CMS, thanks to a huge active community of plug-in developers and theme designers. Joomla allows you to put content in different sections which can be further divided into categories. It has an administration backend where you can configure all settings for core features and installed extensions.

Joomla is very extensible, and supports different types of add-ons which serve different purposes.

Plug-ins: these add features to the site which affect how each page is processed. Such as automatically creating links from wikitag, adding new content editors, or searching content. Plugins are a simple way to add functionality.

Modules: these are like widgets which can be displayed in different sections of the page. For example one could install a polling module to add a poll facility to the website, or add a chatting module to allow people to chat with each other on the site.



Joomla is a popular content management system

except in the most basic of circumstances. However, most CMSs will come with an easy way to create extensions, themes and the like, and it is quite likely that someone has already created the perfect earrings to put on your Mona Lisa.

What are these CMSs like? Let's take two popular examples — Joomla and Drupal.

Components: components are major additions to Joomla. They can sometimes be as complicated as Joomla itself! Adding features such as an image gallery, or a forum, or blogs is something that can be accomplished by the use of components. Major additions to Joomla's core functionality can be made only via components.